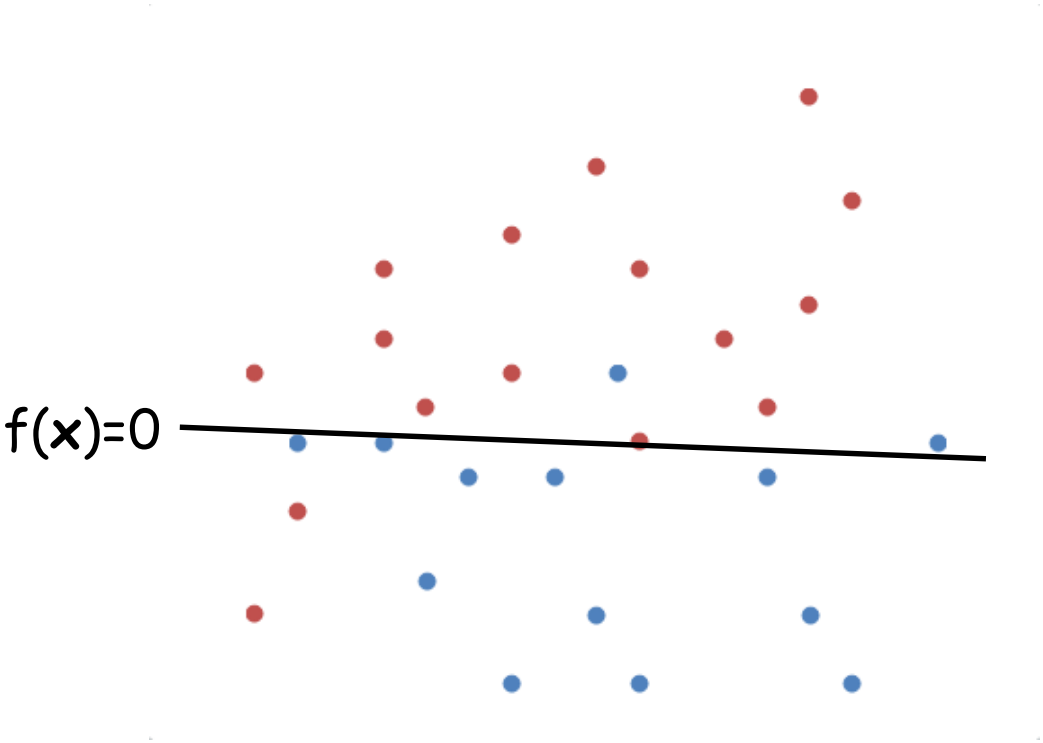


Logistic Regression

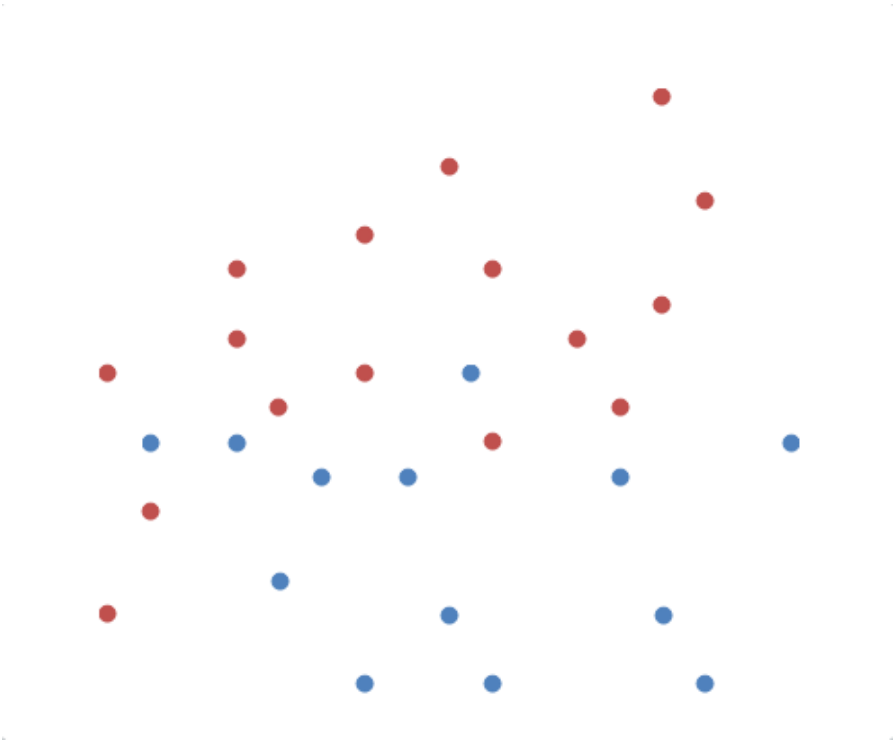
Linear Classifier

- **A linear boundary between classes**



Linear Classifier

- **Find a linear boundary which best discriminates two classes**



Which line?

Linear Classifier

- **Best discrimination: Minimizing error**
  - Error: the number of miss-classified instances
  - So, we need a classifier, *h(x)*, which minimizes

$$\sum_{i=1}^{n} I\left(h(\mathbf{x}_i), y_i\right), \text{ where } I(a,b) = \begin{cases} 1 & a \neq b \\ 0 & a = b \end{cases}$$

or

$$\sum_{i=1}^{n} \left| h(\mathbf{x}_i) - y_i \right|, \text{ since } h(\mathbf{x}_i) \in \{0,1\}, y_i \in \{0,1\}$$

or

$$\sum_{i=1}^{n} \left(h(\mathbf{x}_i) - y_i \right)^2$$

Linear Classifier

- **Find a line of the best discrimination**

Given $D = \{D_1, D_2, \cdots, D_n\}$,

where $D_i = (\mathbf{x}_i, y_i), \mathbf{x}_i = (1, x_1, x_2, \cdots, x_d)$ and $y_i \in \{0,1\}$

Find $\mathbf{w} = (w_0, w_1, \cdots, w_d)$ which minimizes

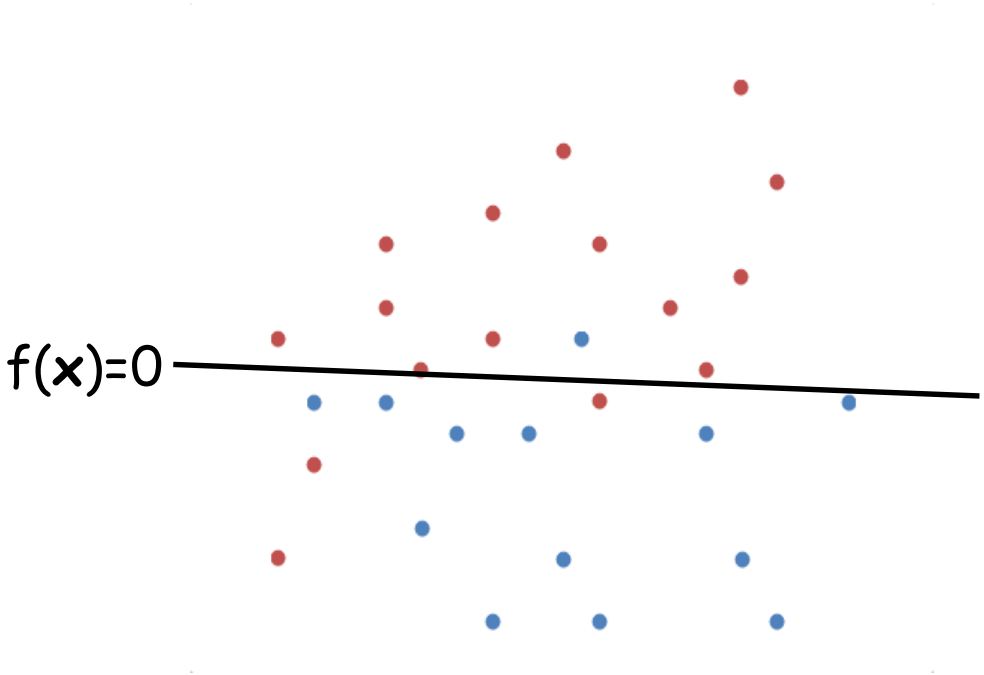
$$E(\mathbf{w}) = \sum_{i=1}^{n} (h(\mathbf{x}_i) - y_i)^2, \text{ where}$$

$$h(\mathbf{x}_i) = \begin{cases} 1 & \text{if } f(\mathbf{x}_i) \geq 0 \\ 0 & \text{otherwise} \end{cases}, \text{ and } f(\mathbf{x}_i) = \sum_{j=0}^{d} w_j x_{ij}$$

  - How to solve this?
    - Not easy to mathematically handle because *h(**x**)* is not differentiable!!

Logistic Regression

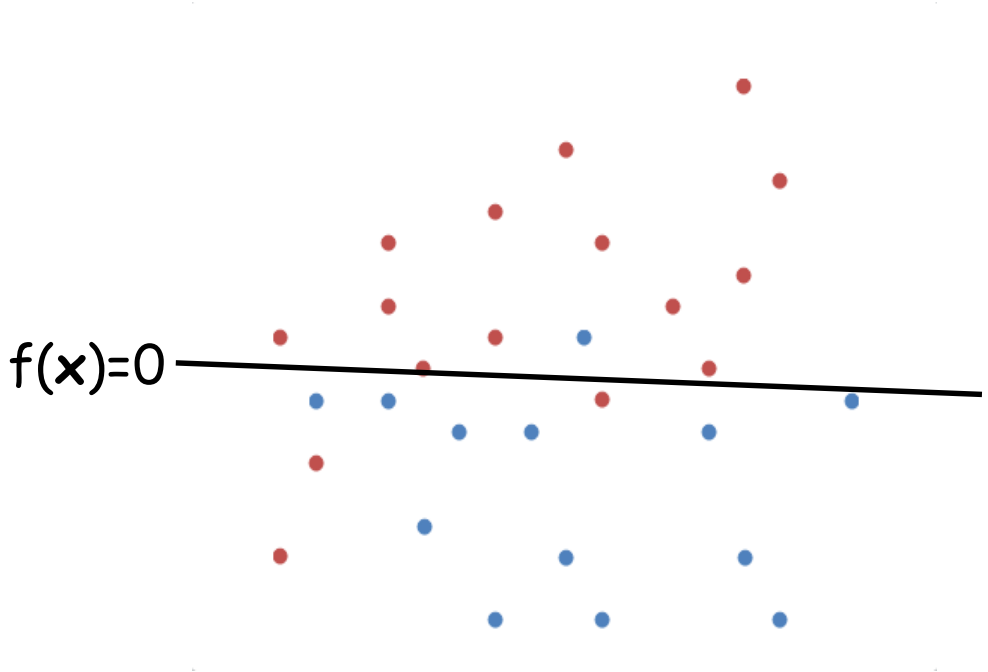
- **Let's modify a little softer**
  - *h(**x**)* can be interpreted as the degree of "being RED"



$$h(\mathbf{x}) = \begin{cases} 1\,(\text{Red}) & \text{if } f(\mathbf{x}) \geq 0 \\ 0\,(\text{Blue}) & \text{if } f(\mathbf{x}) < 0 \end{cases}$$

Logistic Regression

- **Let's modify a little softer**
  - As **x** goes upward from f(**x**), x is more likely to be 1 (RED)
  - As **x** goes downward from f(**x**), x is more likely to be 0 (BLUE)



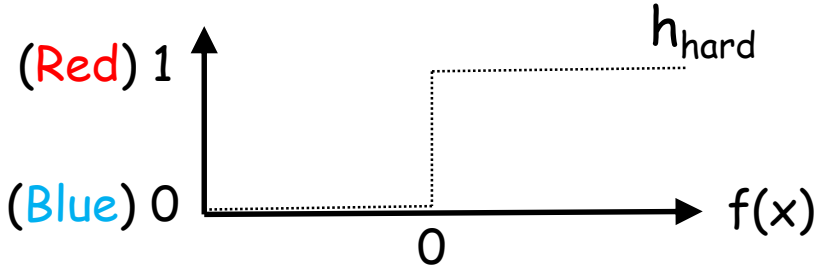
$$h(\mathbf{x}) = \frac{1}{1+\exp\left(-f(\mathbf{x})\right)}$$

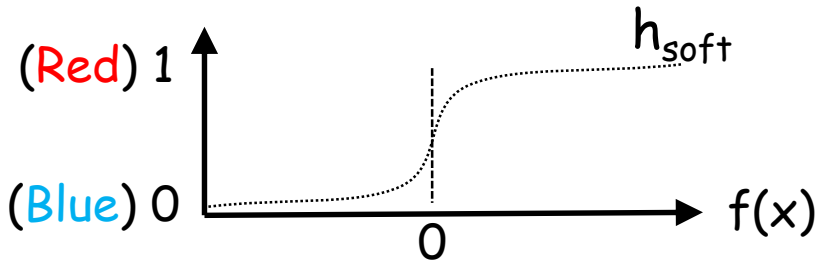
Logistic Regression

- **Soft version of linear classifier**

$$h(\mathbf{x}) = \begin{cases} 1 & \text{if } f(\mathbf{x}) \geq 0 \\ 0 & \text{otherwise} \end{cases} \qquad h(\mathbf{x}) = \frac{1}{1+\exp(-f(\mathbf{x}))}$$

  - If *h(x)* is close to 1, *x* more seems like class 1 (RED)
  - If *h(x)* is close to 0, *x* more seems like class 0 (BLUE)





Logistic Regression

- **Find a *f(x)* of the best discrimination**

Given $D = \{D_1, D_2, \cdots, D_n\}$,

where $D_i = (\mathbf{x}_i, y_i), \mathbf{x}_i = (1, x_1, x_2, \cdots, x_d)$ and $y_i \in \{0,1\}$

Find $\mathbf{w} = (w_0, w_1, \cdots, w_d)$ which minimizes

$$E(\mathbf{w}) = \sum_{i=1}^{n} (h(\mathbf{x}_i) - y_i)^2, \text{ where}$$

$$h(\mathbf{x}_i) = \frac{1}{1 + \exp(-f(\mathbf{x}_i))}, \text{ and } \quad f(\mathbf{x}_i) = \sum_{j=0}^{d} w_j x_{ij}$$

  - How?
    - Using gradient descent method

Logistic Regression

- **Find a *f(x)* of the best discrimination**

$$\frac{\partial E(\mathbf{w})}{\partial w_j} = \frac{1}{2}\sum_{i=1}^{n}\frac{\partial}{\partial w_j}\left(h(\mathbf{x}_i)-y_i\right)^2$$
$$= \frac{1}{2}\sum_{i=1}^{n}\frac{\partial f(\mathbf{x}_i)}{\partial w_j}\frac{\partial h(\mathbf{x}_i)}{\partial f(\mathbf{x}_i)}\frac{\partial}{\partial h(\mathbf{x}_i)}\left(h(\mathbf{x}_i)-y_i\right)^2$$
$$= \sum_{i=1}^{n} x_{ij}h(\mathbf{x}_i)\left(1-h(\mathbf{x}_i)\right)\left(h(\mathbf{x}_i)-y_i\right)$$

$$\frac{\partial f(\mathbf{x}_i)}{\partial w_j} = x_{ij}$$
$$\frac{\partial h(\mathbf{x}_i)}{\partial f(\mathbf{x}_i)} = h(\mathbf{x}_i)\left(1-h(\mathbf{x}_i)\right)$$
$$\frac{\partial}{\partial h(\mathbf{x}_i)}\left(h(\mathbf{x}_i)-y_i\right)^2 = 2\left(h(\mathbf{x}_i)-y_i\right)$$

  - So,
    - Randomly initialize $\mathbf{w}^0 = \left(w_0^0, w_1^0, \cdots, w_d^0\right)$

      Repeat forever for all *j* $w_j^{t+1} = w_j^t - \eta\left.\frac{\partial E(\mathbf{w})}{\partial w_j}\right|_{\mathbf{w}=\mathbf{w}^t}$

Logistic Regression

- **Find a *f(x)* of the best discrimination**

$$w_j^{t+1} = w_j^t - \eta \left. \frac{\partial E(\mathbf{w})}{\partial w_j} \right|_{\mathbf{w}=\mathbf{w}^t}$$

$$\Rightarrow \quad w_j^{t+1} = w_j^t - \eta \sum_{i=1}^{n} x_{ij} h(\mathbf{x}_i)\left(1 - h(\mathbf{x}_i)\right)\left(h(\mathbf{x}_i) - y_i\right)$$

$$\text{where } h(\mathbf{x}_i) = \frac{1}{1 + \exp\left(-f(\mathbf{x}_i)\right)} \quad \text{, and } \quad f(\mathbf{x}_i) = \sum_{j=0}^{d} w_j^t x_{ij}$$

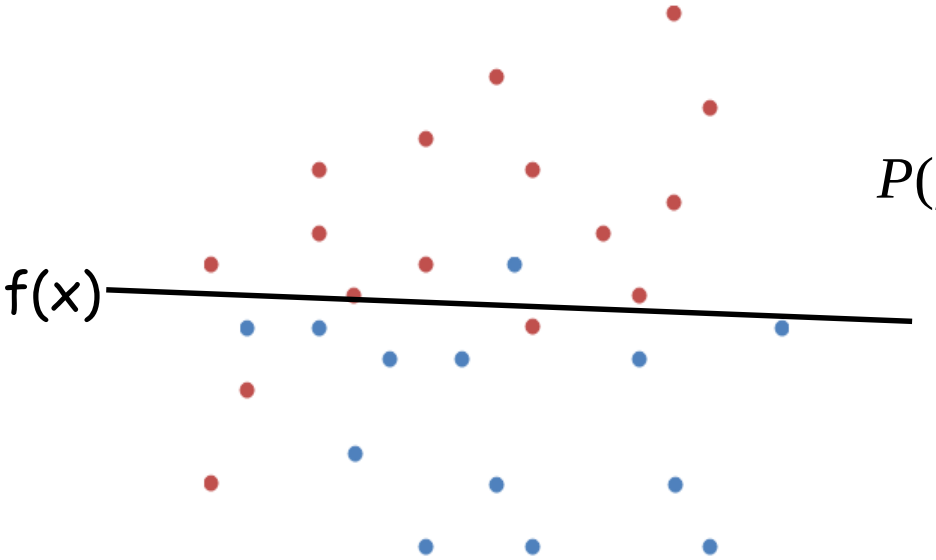
Hmm.. It's not so good.
If x_i is perfectly mis-classified, it is not used to update w's !!

Logistic Regression

- **Find *w* which best fits given data**
  - Approach 1
    - Define an error function
    - Find ***w*** which minimizes the error function
  - Approach 2
    - Evaluate the probability of **D**
    - Find ***w*** which maximizes the probability

Approach with Probability

- **Let's regard h(x) is the probability of "being RED"**
  - As **x** goes upward from f(**x**), x is more likely to be 1 (RED)
  - As **x** goes downward from f(**x**), x is more likely to be 0 (BLUE)

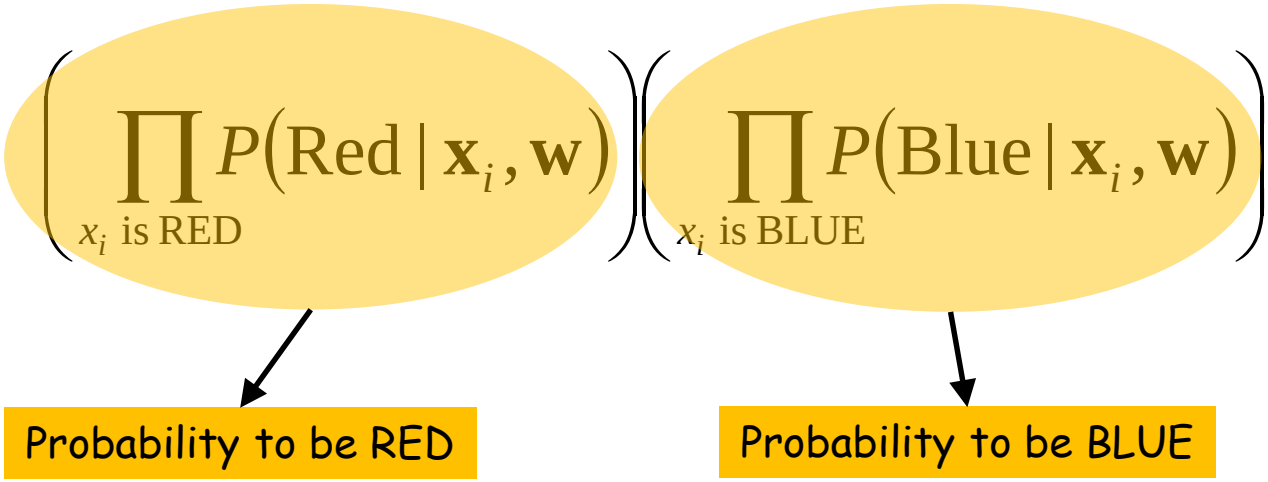


$$P(y_i \mid \mathbf{x}_i, \mathbf{w}) = \begin{cases} h(\mathbf{x}_i) & \text{if } y_i = 1 \\ 1 - h(\mathbf{x}_i) & \text{if } y_i = 0 \end{cases}$$

Approach with Probability

- **Find a boundary which make**
  - Red points have much higher probability to be RED, and
  - Blue points have much higher probability to be BLUE

- **Find a boundary which maximizes**



Approach with Probability

- **Find a boundary**

$$\arg\max_{\mathbf{w}}\left(\prod_{x_i \text{ is RED}} P(\text{Red}\,|\,\mathbf{x}_i,\mathbf{w})\right)\left(\prod_{x_i \text{ is BLUE}} P(\text{Blue}\,|\,\mathbf{x}_i,\mathbf{w})\right)$$

$$= \arg\max_{\mathbf{w}} \log\left(\prod_{i\in\{i|y_i=1\}} h(\mathbf{x}_i)\right)\left(\prod_{i\in\{i|y_i=0\}} 1-h(\mathbf{x}_i)\right)$$

$$= \arg\max_{\mathbf{w}}\left(\sum_{i\in\{i|y_i=1\}} \log h(\mathbf{x}_i) + \sum_{i\in\{i|y_i=0\}} \log\left(1-h(\mathbf{x}_i)\right)\right)$$

$$= \arg\max_{\mathbf{w}}\left(\sum_{i\in\{i|y_i=1\}} \log h(\mathbf{x}_i) + \sum_{i\in\{i|y_i=0\}} \log\left(1-h(\mathbf{x}_i)\right)\right)$$

$$= \arg\max_{\mathbf{w}}\left(\sum_{i\in\{i|y_i=1\}} \left(y_i \log h(\mathbf{x}_i) + (1-y_i)\log\left(1-h(\mathbf{x}_i)\right)\right) + \sum_{i\in\{i|y_i=0\}} \left(y_i \log h(\mathbf{x}_i) + (1-y_i)\log\left(1-h(\mathbf{x}_i)\right)\right)\right)$$

$$= \arg\max_{\mathbf{w}}\left(\sum_{i=1}^{n} \left(y_i \log h(\mathbf{x}_i) + (1-y_i)\log\left(1-h(\mathbf{x}_i)\right)\right)\right)$$

$$P(y_i\,|\,\mathbf{x}_i,\mathbf{w}) = \begin{cases} h(\mathbf{x}_i) & \text{if } y_i = 1\,(\text{Red}) \\ 1-h(\mathbf{x}_i) & \text{if } y_i = 0\,(\text{Blue}) \end{cases}$$

Find *w* which maximize this!!

Approach with Probability

- **Find a boundary**

$$\arg\max_{\mathbf{w}}\left(\sum_{i=1}^{n}\left(y_i \log h(\mathbf{x}_i)+(1-y_i)\log\left(1-h(\mathbf{x}_i)\right)\right)\right)$$

$$=\arg\min_{\mathbf{w}}\left(-\sum_{i=1}^{n}\left(y_i \log h(\mathbf{x}_i)-(1-y_i)\log\left(1-h(\mathbf{x}_i)\right)\right)\right)$$

$$\text{where}\quad h(\mathbf{x})=\frac{1}{1+\exp\left(-\mathbf{w}\cdot\mathbf{x}\right)}$$

  - How to solve it?
    - Gradient Descent Method!!

Approach with Probability

- **Find a boundary by Gradient Descent**

$$E(\mathbf{w}) = -\sum_{i=1}^{n} \left(y_i \log h(\mathbf{x}_i) + (1 - y_i) \log \left(1 - h(\mathbf{x}_i)\right)\right)$$

$$\text{where } h(\mathbf{x}) = g\left(f(\mathbf{x})\right) = \frac{1}{1+\exp\left(-f(\mathbf{x})\right)} = \frac{1}{1+\exp\left(-(w_0 x_0 + w_1 x_1 + \cdots + w_d x_d)\right)}$$

$$\frac{\partial}{\partial w_j} E(\mathbf{w}) = \sum_{i=1}^{n} \left(h(\mathbf{x}_i) - y_i\right) \mathbf{x}_{ij}$$

$$f(\mathbf{x}) = -\mathbf{w}^T \mathbf{x}$$
$$= -(w_0 x_0 + w_1 x_1 + \cdots + w_d x_d)$$
$$\frac{df(\mathbf{x})}{dw_j} = -x_j$$
$$h(x) = \frac{1}{1+\exp(-x)}$$
$$\frac{d}{dx} h(x) = h(x)\left(1 - h(x)\right)$$

Approach with Probability

- **Find a boundary by Gradient Descent**

$$\frac{\partial}{\partial w_j} E(\mathbf{w}) = -\sum_{i=1}^{n} \frac{\partial}{\partial w_j} \left(y_i \log g(f(\mathbf{x}_i)) + (1-y_i) \log \left(1 - g(f(\mathbf{x}_i))\right)\right)$$

$$= -\sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_i)}{\partial w_j} \frac{\partial g(f(\mathbf{x}_i))}{\partial f(\mathbf{x}_i)} \frac{\partial}{\partial g(f(\mathbf{x}_i))} \left(y_i \log g(f(\mathbf{x}_i)) + (1-y_i) \log \left(1 - g(f(\mathbf{x}_i))\right)\right)$$

$$= -\sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_i)}{\partial w_j} \frac{\partial g(f(\mathbf{x}_i))}{\partial f(\mathbf{x}_i)} \left(y_i \frac{1}{g(f(\mathbf{x}_i))} - (1-y_i) \frac{1}{1 - g(f(\mathbf{x}_i))}\right)$$

$$= -\sum_{i=1}^{n} \frac{\partial f(\mathbf{x}_i)}{\partial w_j} g(f(\mathbf{x}_i))\left(1 - g(f(\mathbf{x}_i))\right) \left(y_i \frac{1}{g(f(\mathbf{x}_i))} - (1-y_i) \frac{1}{1 - g(f(\mathbf{x}_i))}\right)$$

$$= -\sum_{i=1}^{n} x_{ij} g(f(\mathbf{x}_i))\left(1 - g(f(\mathbf{x}_i))\right) \left(y_i \frac{1}{g(f(\mathbf{x}_i))} - (1-y_i) \frac{1}{1 - g(f(\mathbf{x}_i))}\right)$$

$$= -\sum_{i=1}^{n} x_{ij} \left(y_i \left(1 - g(f(\mathbf{x}_i))\right) - (1-y_i) g(f(\mathbf{x}_i))\right)$$

$$= -\sum_{i=1}^{n} x_{ij} \left(y_i - g(f(\mathbf{x}_i))\right) = \sum_{i=1}^{n} x_{ij} \left(h(\mathbf{x}_i) - y_i\right)$$

Approach with Probability

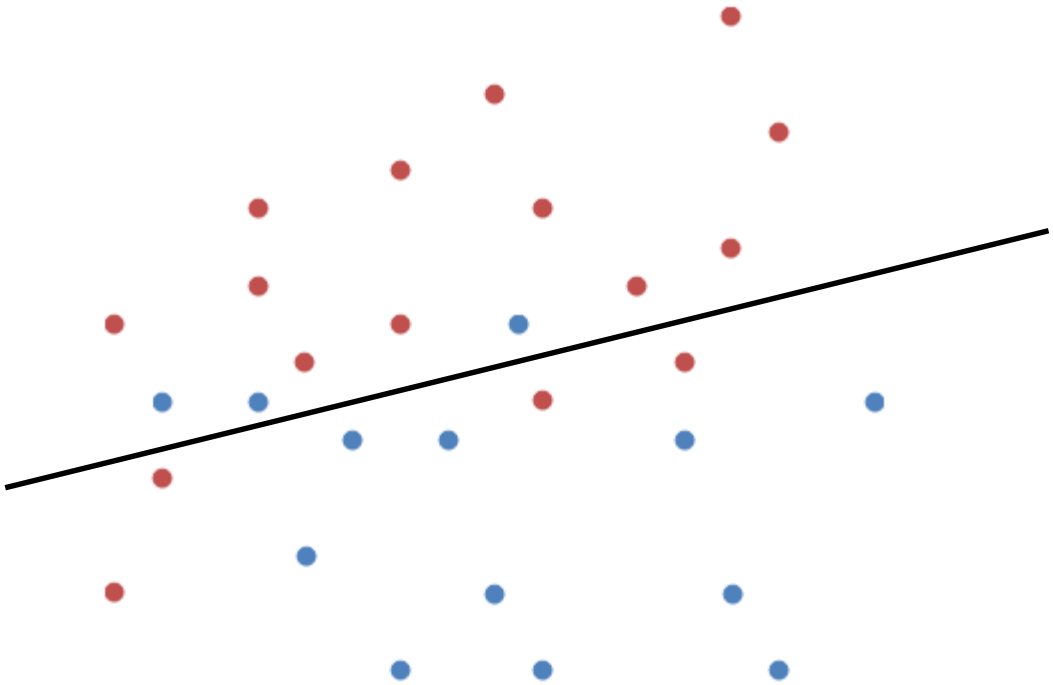
- **How to find a solution of logistic regression**
  - Randomly initialize $\mathbf{w}^0 = \left(w_0^0, w_1^0, \cdots, w_d^0\right)$
  - Repeat forever for all j $w_j^{t+1} = w_j^t - \eta \left.\frac{\partial E(\mathbf{w})}{\partial w_j}\right|_{\mathbf{w}=\mathbf{w}^t}$

$$w_j^{t+1} = w_j^t - \eta \sum_{i=1}^{n} \left(h(\mathbf{x}_i) - y_i\right)\mathbf{x}_{ij}$$

$$\text{where } h(\mathbf{x}_i) = \frac{1}{1+\exp\left(-f(\mathbf{x}_i)\right)} \text{ , and } f(\mathbf{x}_i) = \sum_{j=0}^{d} w_j^t x_{ij}$$

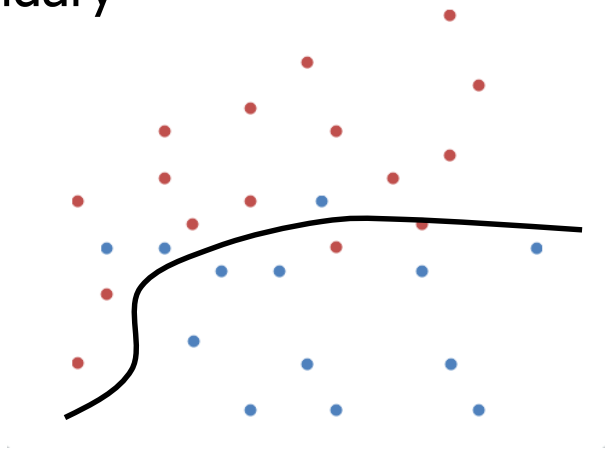
Example

- **Find the boundary**



Discussion

- **Logistic Regression**
  - No closed form solution
    - Optimized by Gradient Descent method
  - A linear boundary



  - Binary classifier
    - If we have more than 3 classes…